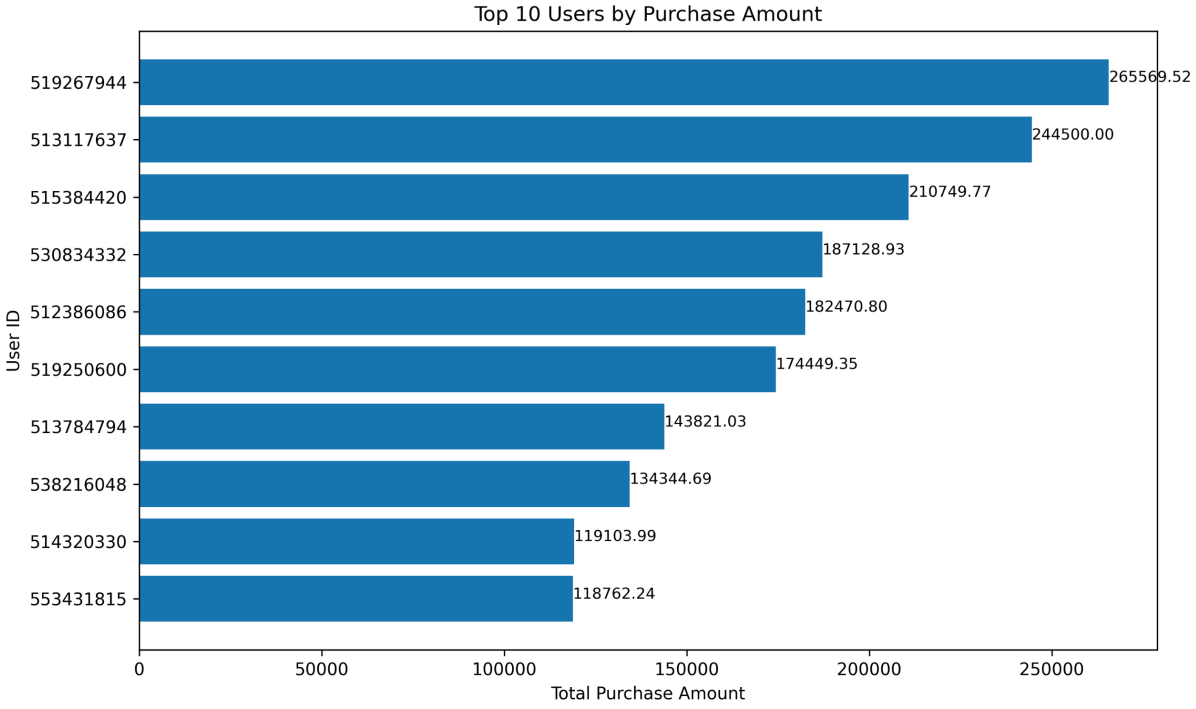
指标分析与用户分群&可视化结果

top10_user_value

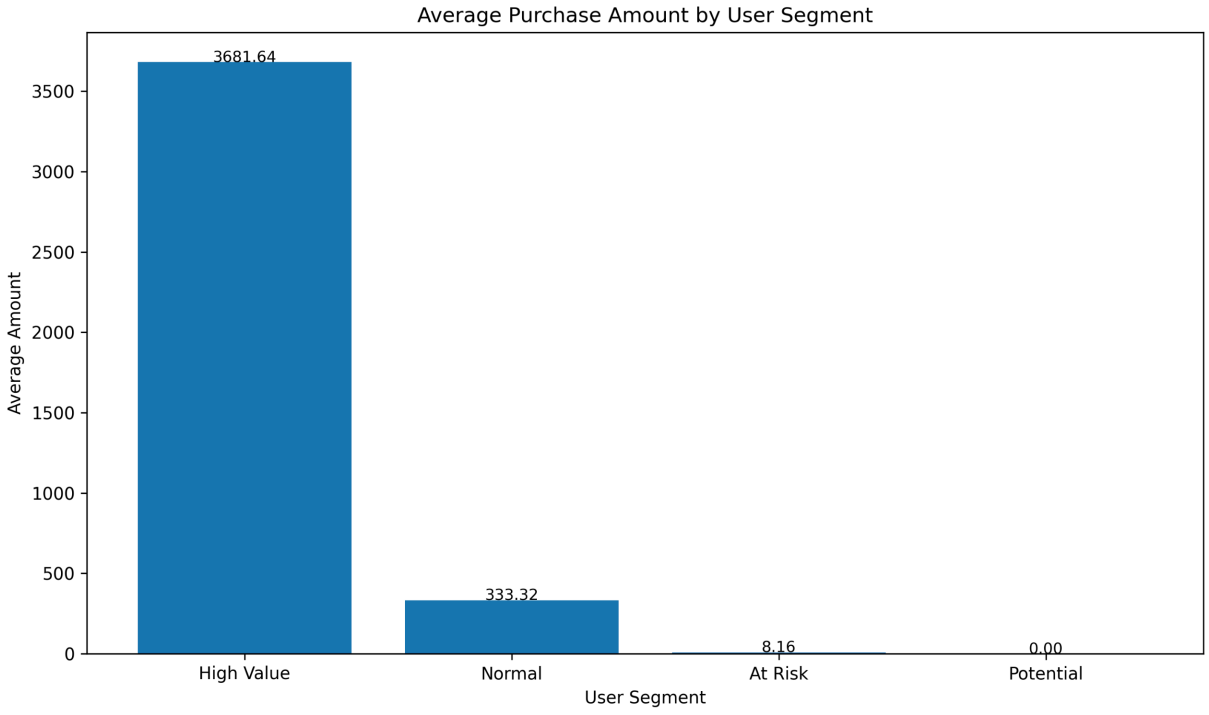
```
hive -e "
use marketing_dw;

insert overwrite local directory '/workspace/marketing_dw/o
utput/n_test/top10_user_value'
row format delimited
fields terminated by ','
select
    user_id,
    total_purchase_amount
from dws_user_behavior_summary
order by total_purchase_amount desc
limit 10;
"
```



> 展示了消费金额排名前10的用户分布情况。可以看出，头部用户的消费金额明显高于普通用户，呈现出典型的“长尾效应”。少数高价值用户贡献了大量销售额，这一现象说明在实际运营中，应重点针对高价值用户制定精细化运营策略，以提升用户生命周期价值

user_segment_avg_amount

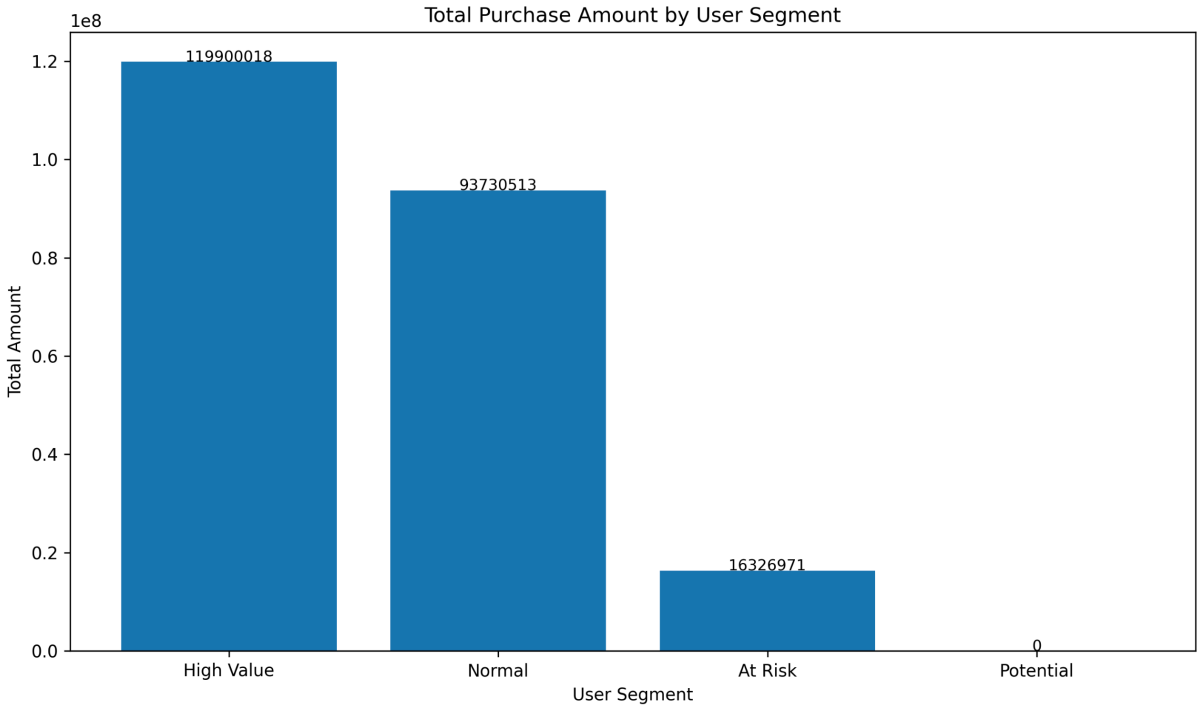
```
hive -e "
use marketing_dw;

insert overwrite local directory '/workspace/marketing_dw/o
utput/n_test/user_segment_avg_amount'
row format delimited
fields terminated by ','
select
    user_value_level,
    avg(total_purchase_amount) as avg_amount
from ads_user_rfm_score
group by user_value_level;
"
```



> 展示了不同用户价值分层的平均消费金额。从结果可以看出，高价值用户的平均消费金额显著高于其他用户群体，表明该类用户具有较强的消费能力和较高的商业价值；潜力用户的平均消费金额处于中等水平，具有进一步转化为高价值用户的潜力；一般用户和流失风险用户的平均消费金额较低，说明其消费活跃度和贡献度相对有限。

user_segment_total_amount

```
hive -e "
use marketing_dw;

insert overwrite local directory '/workspace/marketing_dw/o
utput/n_test/user_segment_total_amount'
row format delimited
fields terminated by ','
select
    user_value_level,
    sum(total_purchase_amount) as total_amount
from ads_user_rfm_score
group by user_value_level;
"
```



> 展示了不同用户分层的总消费金额分布情况。可以看出，高价值用户贡献了绝大部分消费金额，是平台的核心收入来源；潜力用户次之，对整体收入具有一定支撑作用；一般用户和流失风险用户的总消费金额较低，对整体收入贡献有限。因此，在营销策略中应重点关注高价值用户的维护以及潜力用户的转化。

生成以上三张图Python代码

```
cat > /workspace/marketing_dw/output/n_test/plot.py <<'PY'
import os
import pandas as pd
import matplotlib.pyplot as plt

BASE_DIR = "/workspace/marketing_dw/output/n_test"

# ==========================
# 中文 → 英文 映射（核心）
# ==========================
label_map = {
    "高价值用户": "High Value",
    "潜力用户": "Potential",
    "一般用户": "Normal",
    "流失风险用户": "At Risk"
}

# ==========================
# 读取数据
# ==========================
avg_df = pd.read_csv(os.path.join(BASE_DIR, "user_segment_a
vg_amount.csv"),
                     header=None, names=["user_value_leve
l", "avg_amount"]).dropna()

total_df = pd.read_csv(os.path.join(BASE_DIR, "user_segment
_total_amount.csv"),
                       header=None, names=["user_value_leve
l", "total_amount"]).dropna()

top10_df = pd.read_csv(os.path.join(BASE_DIR, "top10_user_v
```

```
alue.csv"),
                        header=None, names=["user_id", "tota
l_purchase_amount"]).dropna()

# ==========================
# 数据类型转换
# ==========================
avg_df["avg_amount"] = pd.to_numeric(avg_df["avg_amount"],
errors="coerce").fillna(0)
total_df["total_amount"] = pd.to_numeric(total_df["total_am
ount"], errors="coerce").fillna(0)
top10_df["total_purchase_amount"] = pd.to_numeric(top10_df
["total_purchase_amount"], errors="coerce").fillna(0)

# ==========================
# 替换中文标签 → 英文
# ==========================
avg_df["user_value_level"] = avg_df["user_value_level"].map
(label_map)
total_df["user_value_level"] = total_df["user_value_leve
l"].map(label_map)

# ==========================
# 排序
# ==========================
avg_df = avg_df.sort_values(by="avg_amount", ascending=Fals
e)
total_df = total_df.sort_values(by="total_amount", ascendin
g=False)
top10_df = top10_df.sort_values(by="total_purchase_amount",
ascending=True)

# ==========================
# 图1: 平均消费
# ==========================
plt.figure(figsize=(10,6))
bars = plt.bar(avg_df["user_value_level"], avg_df["avg_amou
nt"])
```

```
for b in bars:
    plt.text(b.get_x()+b.get_width()/2, b.get_height(),
             f"{b.get_height():.2f}", ha="center", fontsize
=9)

plt.title("Average Purchase Amount by User Segment")
plt.xlabel("User Segment")
plt.ylabel("Average Amount")
plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR, "user_segment_avg_amount
_bar.png"), dpi=300)
plt.close()

# ==========================
# 图2: 总消费
# ==========================
plt.figure(figsize=(10,6))
bars = plt.bar(total_df["user_value_level"], total_df["tota
l_amount"])

for b in bars:
    plt.text(b.get_x()+b.get_width()/2, b.get_height(),
             f"{b.get_height():.0f}", ha="center", fontsize
=9)

plt.title("Total Purchase Amount by User Segment")
plt.xlabel("User Segment")
plt.ylabel("Total Amount")
plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR, "user_segment_total_amou
nt_bar.png"), dpi=300)
plt.close()

# ==========================
# 图3: Top10用户
# ==========================
plt.figure(figsize=(10,6))
```

```
bars = plt.barh(top10_df["user_id"].astype(str),
                top10_df["total_purchase_amount"])

for b in bars:
    plt.text(b.get_width(), b.get_y()+b.get_height()/2,
             f"{b.get_width():.2f}", fontsize=9)

plt.title("Top 10 Users by Purchase Amount")
plt.xlabel("Total Purchase Amount")
plt.ylabel("User ID")
plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR, "top10_user_value_barh.p
ng"), dpi=300)
plt.close()

print("All charts generated successfully (English versio
n)")
PY
```

导出各分层用户人数

```
hive -e "
use marketing_dw;

insert overwrite local directory '/workspace/marketing_dw/o
utput/n_test/user_segment_count'
row format delimited
fields terminated by ','
select
    user_value_level,
    count(*) as user_cnt
from ads_user_rfm_score
group by user_value_level;
"
```

导出各分层平均行为指标

`ads_user_rfm_score` 和 `dws_user_behavior_summary` 关联，按用户分层统计

```
hive -e "
use marketing_dw;

insert overwrite local directory '/workspace/marketing_dw/o
utput/n_test/user_segment_behavior_metrics'
row format delimited
fields terminated by ','
select
    a.user_value_level,
    avg(b.total_view_cnt) as avg_view_cnt,
    avg(b.total_cart_cnt) as avg_cart_cnt,
    avg(b.total_purchase_cnt) as avg_purchase_cnt
from ads_user_rfm_score a
join dws_user_behavior_summary b
on a.user_id = b.user_id
group by a.user_value_level;
"
```

导出各分层平均活跃天数

```
hive -e "
use marketing_dw;

insert overwrite local directory '/workspace/marketing_dw/o
utput/n_test/user_segment_active_days'
row format delimited
fields terminated by ','
select
    a.user_value_level,
    avg(b.active_day_cnt) as avg_active_day_cnt
from ads_user_rfm_score a
join dws_user_behavior_summary b
```

```
on a.user_id = b.user_id
group by a.user_value_level;
"
```

导出各分层购买转化率

purchase_conversion_rate = sum(total_purchase_cnt) / sum(total_view_cnt)

```
hive -e "
use marketing_dw;

insert overwrite local directory '/workspace/marketing_dw/o
utput/n_test/user_segment_conversion_rate'
row format delimited
fields terminated by ','
select
    a.user_value_level,
    case
        when sum(b.total_view_cnt) = 0 then 0
        else sum(b.total_purchase_cnt) / sum(b.total_view_c
nt)
    end as purchase_conversion_rate
from ads_user_rfm_score a
join dws_user_behavior_summary b
on a.user_id = b.user_id
group by a.user_value_level;
"
```

以上四张图片可视化脚本

```
cat > /workspace/marketing_dw/output/n_test/plot_advanced.p
y <<'PY'
import os
import pandas as pd
import matplotlib.pyplot as plt
```

```
BASE_DIR = "/workspace/marketing_dw/output/n_test"

label_map = {
    "高价值用户": "High Value",
    "潜力用户": "Potential",
    "一般用户": "Normal",
    "流失风险用户": "At Risk"
}

segment_order = ["High Value", "Potential", "Normal", "At R
isk"]

# ========================
# 读取数据
# ========================
count_df = pd.read_csv(
    os.path.join(BASE_DIR, "user_segment_count.csv"),
    header=None,
    names=["user_value_level", "user_cnt"]
).dropna()

behavior_df = pd.read_csv(
    os.path.join(BASE_DIR, "user_segment_behavior_metrics.c
sv"),
    header=None,
    names=["user_value_level", "avg_view_cnt", "avg_cart_cn
t", "avg_purchase_cnt"]
).dropna()

active_df = pd.read_csv(
    os.path.join(BASE_DIR, "user_segment_active_days.csv"),
    header=None,
    names=["user_value_level", "avg_active_day_cnt"]
).dropna()

conversion_df = pd.read_csv(
    os.path.join(BASE_DIR, "user_segment_conversion_rate.cs
v"),
```

```
    header=None,
    names=["user_value_level", "purchase_conversion_rate"]
).dropna()

# ========================
# 数据转换
# ========================
count_df["user_cnt"] = pd.to_numeric(count_df["user_cnt"],
errors="coerce").fillna(0)
behavior_df["avg_view_cnt"] = pd.to_numeric(behavior_df["av
g_view_cnt"], errors="coerce").fillna(0)
behavior_df["avg_cart_cnt"] = pd.to_numeric(behavior_df["av
g_cart_cnt"], errors="coerce").fillna(0)
behavior_df["avg_purchase_cnt"] = pd.to_numeric(behavior_df
["avg_purchase_cnt"], errors="coerce").fillna(0)
active_df["avg_active_day_cnt"] = pd.to_numeric(active_df
["avg_active_day_cnt"], errors="coerce").fillna(0)
conversion_df["purchase_conversion_rate"] = pd.to_numeric(c
onversion_df["purchase_conversion_rate"], errors="coerce").
fillna(0)

# 中文转英文
count_df["user_value_level"] = count_df["user_value_leve
l"].map(label_map)
behavior_df["user_value_level"] = behavior_df["user_value_l
evel"].map(label_map)
active_df["user_value_level"] = active_df["user_value_leve
l"].map(label_map)
conversion_df["user_value_level"] = conversion_df["user_val
ue_level"].map(label_map)

# 固定顺序
count_df["user_value_level"] = pd.Categorical(count_df["use
r_value_level"], categories=segment_order, ordered=True)
behavior_df["user_value_level"] = pd.Categorical(behavior_d
f["user_value_level"], categories=segment_order, ordered=Tr
ue)
active_df["user_value_level"] = pd.Categorical(active_df["u
```

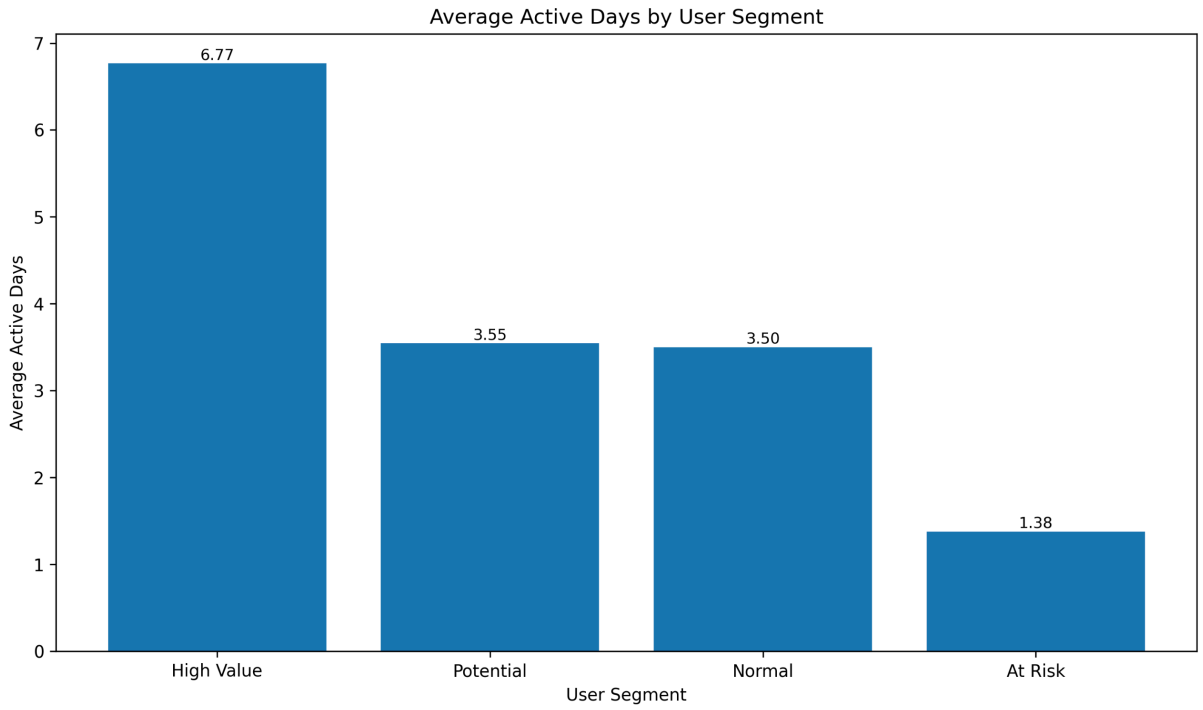
```
ser_value_level"], categories=segment_order, ordered=True)
conversion_df["user_value_level"] = pd.Categorical(conversi
on_df["user_value_level"], categories=segment_order, ordere
d=True)

count_df = count_df.sort_values("user_value_level")
behavior_df = behavior_df.sort_values("user_value_level")
active_df = active_df.sort_values("user_value_level")
conversion_df = conversion_df.sort_values("user_value_leve
l")

# ==========================
# 图1：用户分层人数占比饼图
# ==========================
plt.figure(figsize=(8, 8))
plt.pie(
    count_df["user_cnt"],
    labels=count_df["user_value_level"],
    autopct="%1.1f%%",
    startangle=90
)
plt.title("User Segment Distribution")
plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR, "user_segment_count_pie.
png"), dpi=300)
plt.close()

# ==========================
# 图2：平均行为强度分组柱状图
# ==========================
x = range(len(behavior_df))
width = 0.25

plt.figure(figsize=(10, 6))
plt.bar([i - width for i in x], behavior_df["avg_view_cn
t"], width=width, label="Avg Views")
plt.bar(x, behavior_df["avg_cart_cnt"], width=width, label
="Avg Carts")
```

```
plt.bar([i + width for i in x], behavior_df["avg_purchase_c
nt"], width=width, label="Avg Purchases")

plt.xticks(list(x), behavior_df["user_value_level"])
plt.title("Average User Behavior by Segment")
plt.xlabel("User Segment")
plt.ylabel("Average Count")
plt.legend()
plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR, "user_segment_behavior_g
rouped_bar.png"), dpi=300)
plt.close()

# ==========================
# 图3：平均活跃天数柱状图
# ==========================
plt.figure(figsize=(10, 6))
bars = plt.bar(active_df["user_value_level"], active_df["av
g_active_day_cnt"])

for b in bars:
    h = b.get_height()
    plt.text(b.get_x() + b.get_width() / 2, h, f"{h:.2f}",
ha="center", va="bottom", fontsize=9)

plt.title("Average Active Days by User Segment")
plt.xlabel("User Segment")
plt.ylabel("Average Active Days")
plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR, "user_segment_active_day
s_bar.png"), dpi=300)
plt.close()

# ==========================
# 图4：购买转化率柱状图
# ==========================
plt.figure(figsize=(10, 6))
bars = plt.bar(conversion_df["user_value_level"], conversio
```

```
n_df["purchase_conversion_rate"])

for b in bars:
    h = b.get_height()
    plt.text(b.get_x() + b.get_width() / 2, h, f"{h:.4f}",
ha="center", va="bottom", fontsize=9)

plt.title("Purchase Conversion Rate by User Segment")
plt.xlabel("User Segment")
plt.ylabel("Conversion Rate")
plt.tight_layout()
plt.savefig(os.path.join(BASE_DIR, "user_segment_conversion
_rate_bar.png"), dpi=300)
plt.close()

print("Advanced charts generated successfully:")
print("user_segment_count_pie.png")
print("user_segment_behavior_grouped_bar.png")
print("user_segment_active_days_bar.png")
print("user_segment_conversion_rate_bar.png")
PY
```

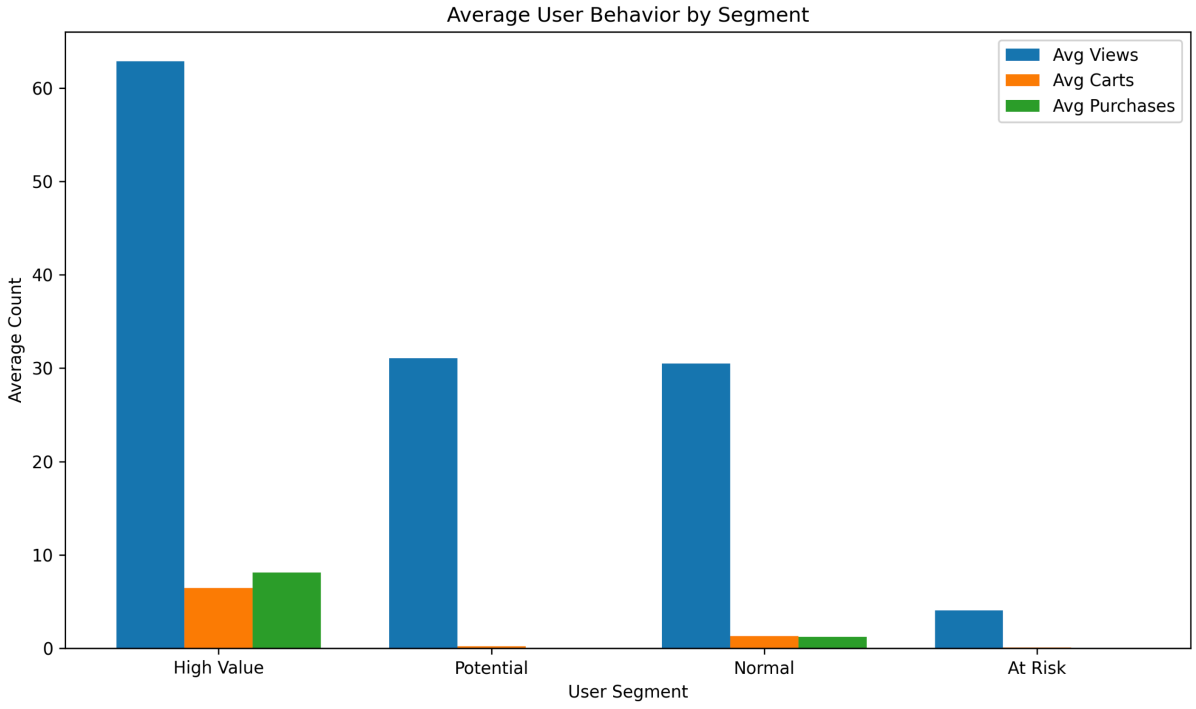
python3 /workspace/marketing_dw/output/n_test/plot_advanced.py

user_segment_active_days_bar.png



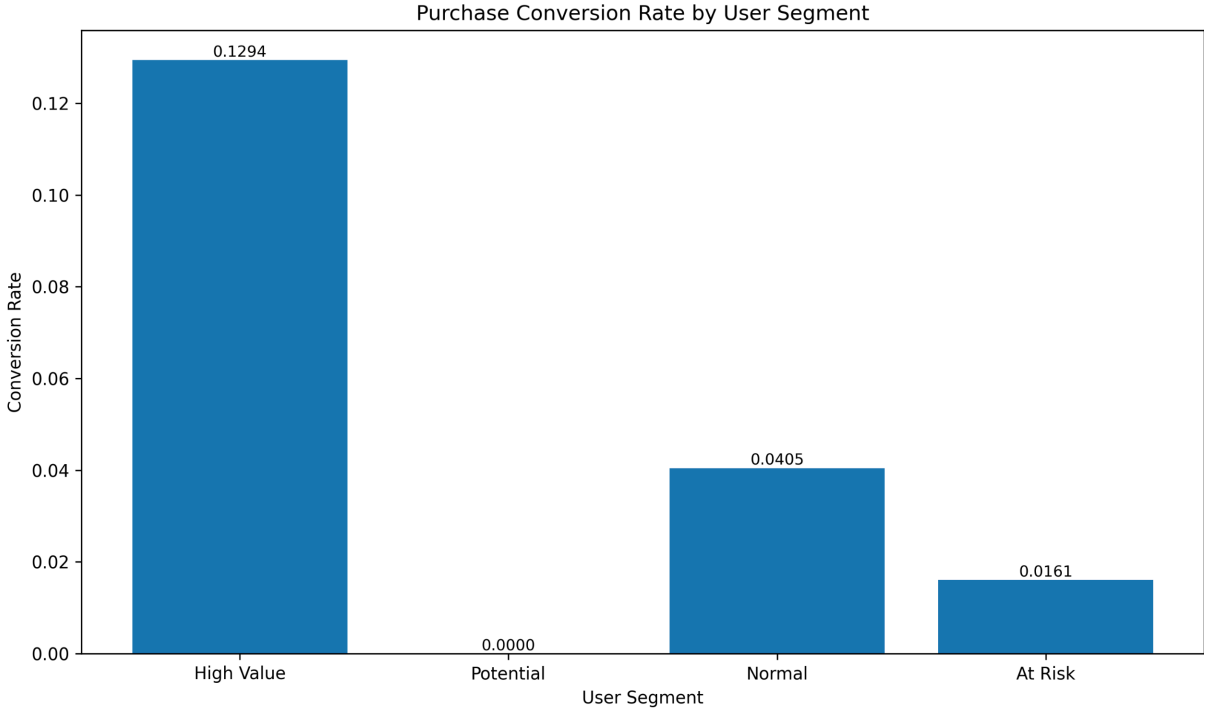
> 展示了不同用户分层的平均活跃天数情况。结果表明，高价值用户的活跃天数显著高于其他用户群体，说明其具有更高的用户粘性和持续参与度；潜力用户和一般用户的活跃程度居中；流失风险用户的活跃天数最低，反映出其流失趋势明显

user_segment_behavior_grouped_bar.png



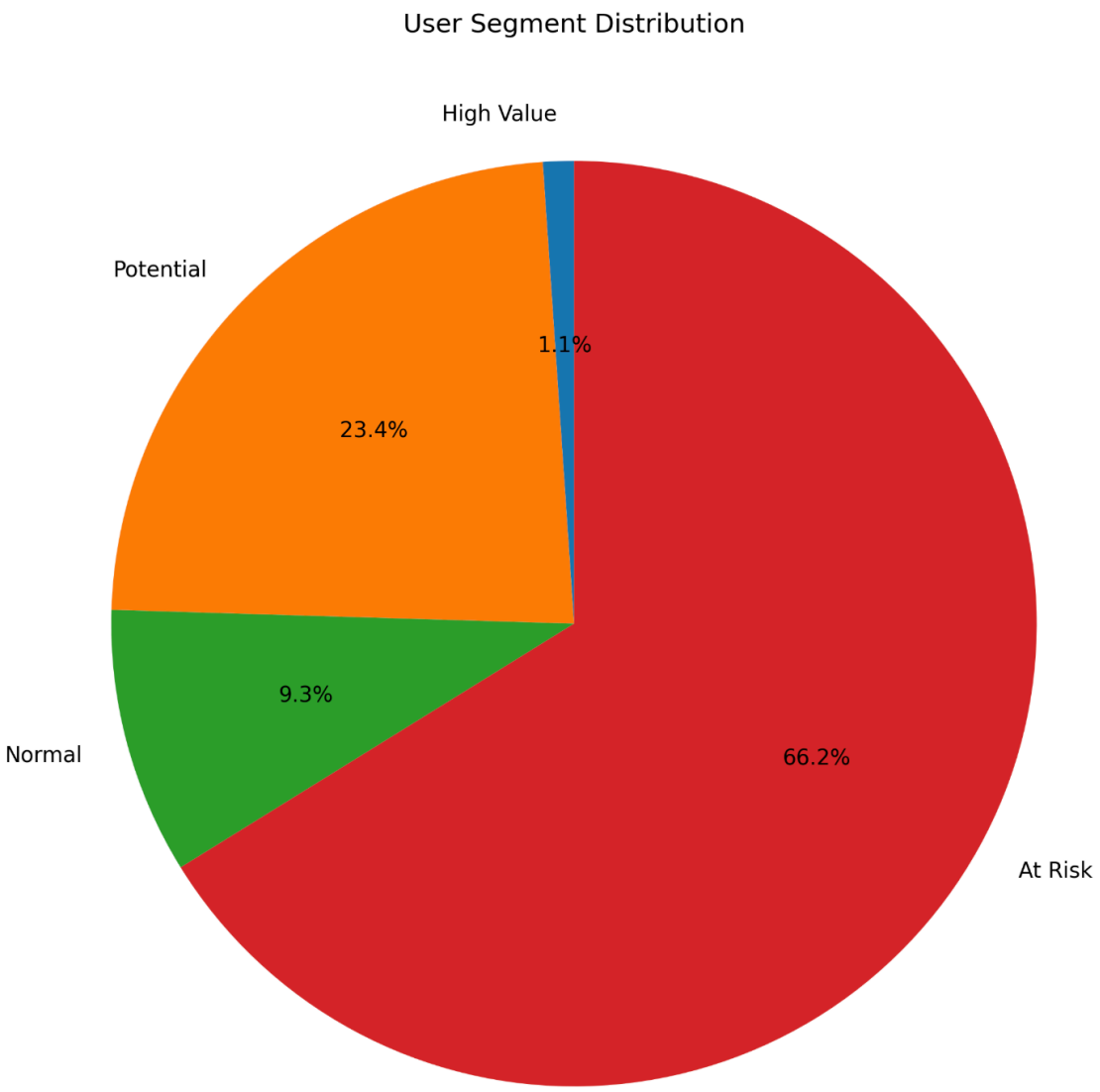
> 展示了不同用户分层在浏览、加购和购买行为上的平均次数差异。可以看出，高价值用户在各类行为指标上均表现活跃，尤其是在购买行为上明显高于其他用户群体；潜力用户具有较高的浏览行为但购买转化不足；流失风险用户整体行为活跃度较低。该结果验证了用户分层在行为特征上的显著差异性。

user_segment_conversion_rate_bar.png



> 展示了不同用户分层的购买转化率差异。可以看出，高价值用户具有最高的转化率，说明其购买意愿强烈；潜力用户虽然浏览行为较多，但转化率相对较低，具有较大的提升空间；流失风险用户的转化率接近于零，表明其已基本处于非活跃状态。该结果为后续精准营销策略提供了重要依据。

user_segment_count_pie.png



> 展示了不同用户价值分层的用户数量占比情况。从图中可以看出，一般用户和流失风险用户占比较高，而高价值用户占比较低但价值突出。这种分布符合典型的用户金字塔结构，说明平台需要通过精细化运营不断提升普通用户向高价值用户的转化比例